

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

software Engineering [NPT-4012]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How do Agile and Waterfall methodologies differ in terms of risk management? (5)
 - A)
 - B) Is it possible to combine software process models for a given project? Justify with suitable example. (5)
- 2) Why don't many software developers pay enough attention to requirements engineering? Are there ever circumstances where you can skip it? Justify your answer. (5)
 - A)
 - B) What strategies can help manage conflicting requirements during validation? (3)
 - C) What is the need for design patterns? Discuss any 2 design patterns with suitable example. (2)
- 3) You are assigned as a business analyst for a software project to develop an online learning platform. Conduct a requirement analysis for this project by identifying and categorizing at least five functional and five non-functional requirements. Discuss the importance of each category in the context of the project. (5)
 - A)
 - B) Illustrate with neat sketches, advantages and disadvantages of the Pipes and Filters and layered architectural design models. (3)
 - C) Compare and contrast software architecture and software design. (2)
- 4) Design the Use case diagrams, class diagrams and sequence diagrams for MIT Sports Management System. (5)
 - A)
 - B) What elements of a user interface design can be evaluated without building a working computer prototype? (3)
 - C) Illustrate any five differences between software test engineer and hardware test engineer. (2)
- 5) Illustrate various defects encountered at different phases of software life cycle. (5)

- A)
- B) Design the Control-flow graph for the Triangle problem and compute the cyclomatic complexity for the same. (3)
- C) Distinguish between:
 - 1. Alpha testing and Beta Testing (2)
 - 2. Performance testing and stress testing

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